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Experiment 2: RF Components: Characterization and Measurement

1 Aim

To understand the operation and experimentally characterize the filter, mixer, power splitter/combiner, and amplifier used in an RF receiver chain.

This report covers **Part A: Filter characterization** (A1: transmission response and insertion loss; A2: filter input reflection coefficient and return loss; A3: quarter-wave open-circuit stub notch filter).

2 Objectives

The objectives covered in this report are:

(a) To characterize an RF filter, identify its type, and determine its cutoff frequencies or passband.

(b) To demonstrate a quarter-wave open-circuit stub as a notch filter.

(f) To estimate the input reflection coefficient and return loss of the filter.

3 Theory

3.1 Filter transmission characteristics

A filter passes selected frequency components and attenuates others. The four commonly used types are low-pass, high-pass, band-pass, and band-reject (notch) filters. For a low-pass or high-pass filter, f_c denotes the cutoff frequency and f_s the stopband edge; for band-pass and notch filters, f_1 and f_2 denote the lower and upper cutoff frequencies and f_0 the centre or notch frequency.

Let P_{in} be the power measured without the filter and P_{out} the power measured after inserting the filter. The insertion loss is

$$\text{IL} = P_{\text{in}} - P_{\text{out}} \ \text{ dB}. \tag{1}$$

A smaller IL indicates better transmission. The -3 dB cutoff frequencies are measured relative to the minimum insertion loss in the passband, i.e. at

$$\text{IL}(f_c) = \text{IL}_{\text{min}} + 3 \text{ dB}. \tag{2}$$

3.2 Filter reflection coefficient and return loss

If the filter input impedance Z_{in} differs from the characteristic impedance Z_0 of the RF system, part of the incident signal is reflected. The voltage reflection coefficient is

$$\Gamma = \frac{V_{\text{ref}}}{V_{\text{inc}}} = \frac{Z_{\text{in}} - Z_0}{Z_{\text{in}} + Z_0}, \tag{3}$$

where V_{inc} and V_{ref} are the incident and reflected voltage amplitudes. The return loss is

$$\text{RL} = -20 \log_{10} |\Gamma| \ \text{ dB}. \tag{4}$$

The limiting cases are:

Condition	$\lvert\Gamma\rvert$	RL
Perfect match ($Z_{\text{in}} = Z_0$)	0	∞
Partial mismatch	$0 < \lvert\Gamma\rvert < 1$	> 0 dB
Complete reflection	1	0 dB

In the Tee-junction method the oscilloscope measures, at the Tee, the superposition of the incident and reflected waves, $V_{\text{osc}} = V_{\text{incident}} + V_{\text{reflected}}$. As the frequency is swept, the relative phase of the two waves changes, so the amplitude varies between

$$V_{\max} = |V_{\text{inc}}| + |V_{\text{ref}}|, \qquad V_{\min} = |V_{\text{inc}}| - |V_{\text{ref}}|. \tag{5}$$

Adding and subtracting these gives

$$V_{\text{inc}} = \frac{V_{\max} + V_{\min}}{2}, \qquad V_{\text{ref}} = \frac{V_{\max} - V_{\min}}{2}, \tag{6}$$

and therefore

$$|\Gamma| = \frac{V_{\text{ref}}}{V_{\text{inc}}} = \frac{V_{\max} - V_{\min}}{V_{\max} + V_{\min}}. \tag{7}$$

3.3 Quarter-wave open-circuit stub filter

An open-ended coaxial stub connected in shunt through a Tee connector behaves as a resonator. At its quarter-wave resonant frequency, the open circuit at the far end is transformed approximately into a low impedance at the Tee junction, so the signal is strongly attenuated and a notch is produced. The stub length is

$$L = \frac{\lambda}{4} = \frac{\text{VF} \cdot c}{4 f_0}, \qquad \text{so} \qquad f_0 = \frac{\text{VF} \cdot c}{4L}, \tag{8}$$

where λ is the guided wavelength in the cable, VF is the velocity factor, $c \approx 3 \times 10^8$ m/s is the speed of light in vacuum, and f_0 is the notch frequency. The connector length and open-end effect may cause the measured f_0 to differ slightly from the calculated value.

4 Components Required (Part A)

RF signal generator; spectrum analyser; digital storage oscilloscope; filter; RF Tee connector; open-ended coaxial stub; short reference coaxial cable; 50 Ω termination; suitable RF cables and adapters.

5 Experimental Setup and Precautions

A1 (Setup A). The RF signal generator is connected to the device under test (reference cable or filter) and the output is measured with the spectrum analyser.

A2 (Setup B). The signal generator is connected to one port of the Tee connector and the oscilloscope to the second port. The filter input is connected to the third port, and the filter output is terminated with a 50 Ω load. The oscilloscope input impedance is set to 1 MΩ so that the oscilloscope does not significantly load the Tee junction.

A3. The signal generator is connected to one port of a Tee connector and the spectrum analyser to the second port. An open-ended coaxial stub of length L is connected to the third port.

Precautions.

- Generator power is kept constant when comparing the reference connection with the device-under-test measurement.
- The same cables and adapters are used for reference and device measurements.
- The operating-frequency range and maximum safe input power of every component, and the spectrum analyser input limit, are not exceeded.
- For A2, the frequency sweep is restricted to a small range within a relatively flat part of the filter passband, with constant generator amplitude.

6 Procedure

6.1 A1: Transmission response and insertion loss

1. The signal generator was connected to the spectrum analyser through the short reference cable and the generator power was set to a constant value.
2. P_{in} was measured without the filter.
3. The filter was inserted without changing the generator setting, cables, or adapters, and P_{out} was measured at the same frequencies.
4. $\text{IL} = P_{\text{in}} - P_{\text{out}}$ was calculated (Eq. 1) and plotted versus frequency; the filter type and the relevant frequencies were identified.

6.2 A2: Filter input reflection coefficient and return loss

1. The connections of Setup B were made and the oscilloscope input impedance was set to 1 MΩ.
2. A small frequency range within a flat part of the passband was selected and swept slowly at constant generator amplitude.
3. $V_{\max}$ and the adjacent $V_{\min}$ at the Tee junction were recorded.
4. V_{inc}, V_{ref}, $|\Gamma|$ and RL were calculated using Eqs. 6, 7 and 4.

6.3 A3: Quarter-wave open-circuit stub notch filter

1. The generator, spectrum analyser and open-ended stub were connected to the three ports of the Tee connector.
2. The expected notch frequency $f_0 = \text{VF} \cdot c/(4L)$ was calculated (Eq. 8).
3. The notch frequency was measured with the spectrum analyser and compared with the calculated value.

7 Observations

7.1 A1: Filter transmission

Generator power / input power: $P_{\mathrm{in}} = 0\,\mathrm{dBm}$ (recorded at 50 MHz and taken as the same at all frequencies). Cable loss recorded at 50 MHz: 0.61 dB (not subtracted; IL is calculated from Eq. 1).

f (MHz)	P_{in} (dBm)	P_{out} (dBm)	IL (dB)	f (MHz)	P_{in} (dBm)	P_{out} (dBm)	IL (dB)
50	0	−0.8	0.8	580	0	−6.4	6.4
100	0	−0.88	0.88	590	0	−7.4	7.4
150	0	−1.02	1.02	600	0	−8.01	8.01
200	0	−1.1	1.1	610	0	−8.5	8.5
250	0	−1.46	1.46	620	0	−9.35	9.35
300	0	−1.73	1.73	630	0	−11.5	11.5
350	0	−1.44	1.44	640	0	−15.2	15.2
400	0	−2.03	2.03	650	0	−20	20
410	0	−1.85	1.85	660	0	−24	24
420	0	−1.76	1.76	670	0	−26	26
430	0	−1.88	1.88	680	0	−27	27
440	0	−2.12	2.12	690	0	−28.5	28.5
450	0	−2.33	2.33	700	0	−31	31
460	0	−2.43	2.43	710	0	−35.4	35.4
470	0	−2.43	2.43	720	0	−38	38
480	0	−2.43	2.43	730	0	−41	41
490	0	−2.55	2.55	740	0	−42	42
500	0	−2.81	2.81	750	0	−41	41
510	0	−3.25	3.25	760	0	−39	39
520	0	−3.6	3.6	770	0	−38	38
530	0	−3.77	3.77	780	0	−38	38
540	0	−3.8	3.8	790	0	−38	38
550	0	−3.84	3.84	800	0	−38	38
560	0	−4.3	4.3	850	0	−34	34
570	0	−5.2	5.2	900	0	−36	36

Table 1: Filter transmission measurement: P_{in} (without filter), P_{out} (with filter) and IL = $P_{\mathrm{in}} - P_{\mathrm{out}}$.

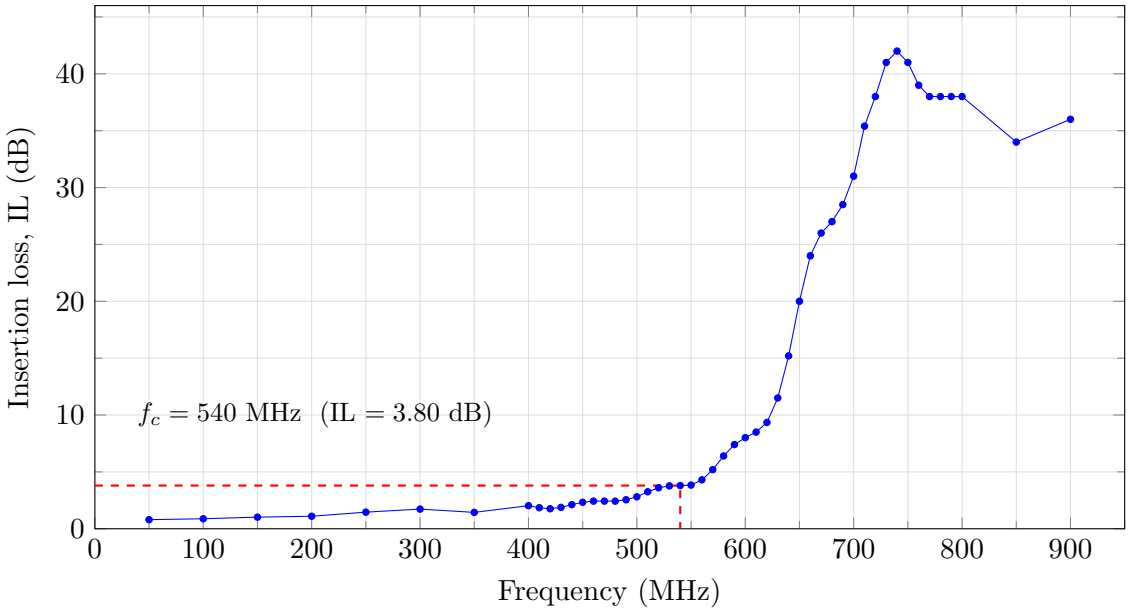


Figure 1: Insertion loss of the filter versus frequency. The dashed lines mark the −3 dB point relative to the minimum passband insertion loss.

7.2 A2: Filter input reflection coefficient and return loss

Frequency range	$V_{\max}$ (mV)	$V_{\min}$ (mV)	V_{inc} (mV)	V_{ref} (mV)	RL (dB)
60–65 MHz	680 (at 65 MHz)	645 (at 60 MHz)	662.5	17.5	31.56

Table 2: Tee-junction measurement for the filter input in the passband.

7.3 A3: Quarter-wave open-circuit stub

Quantity	Symbol	Value
Stub length	L	64 cm
Velocity factor used	VF	0.66
Calculated notch frequency	f_0 (calc.)	77.3 MHz
Measured notch frequency (spectrum analyser)	f_0 (meas.)	59 MHz

Table 3: Stub parameters and notch frequencies.

8 Calculations

8.1 A1: Insertion loss and cutoff frequency

From Eq. 1, with $P_{\text{in}} = 0$ dBm:

$$\begin{aligned} f = 50 \text{ MHz}: &\quad \text{IL} = 0 - (-0.8) = 0.80 \text{ dB}, \\ f = 540 \text{ MHz}: &\quad \text{IL} = 0 - (-3.8) = 3.80 \text{ dB}. \end{aligned}$$

The minimum insertion loss in the measured passband is $\mathrm{IL}_{\min} = 0.80$ dB (at 50 MHz). From Eq. 2, the -3 dB cutoff occurs at

$$\mathrm{IL}(f_c) = 0.80 + 3 = 3.80 \text{ dB},$$

which is reached at $f_c = 540$ MHz (Table 1; IL = 3.77 dB at 530 MHz and 3.84 dB at 550 MHz). The maximum insertion loss measured is 42 dB at 740 MHz.

8.2 A2: Reflection coefficient and return loss

From Eq. 6:

$$V_{\mathrm{inc}} = \frac{680 + 645}{2} = 662.5 \text{ mV},$$
$$V_{\mathrm{ref}} = \frac{680 - 645}{2} = 17.5 \text{ mV}.$$

From Eq. 7:

$$|\Gamma| = \frac{17.5}{662.5} = \frac{680 - 645}{680 + 645} = 0.0264.$$

From Eq. 4:

$$\mathrm{RL} = -20 \log_{10}(0.0264) = 31.56 \text{ dB}.$$

8.3 A3: Stub notch frequency

From Eq. 8 with VF $= 0.66$, $c = 3 \times 10^8$ m/s and $L = 0.64$ m:

$$f_0 = \frac{0.66 \times 3 \times 10^8}{4 \times 0.64} = \frac{1.98 \times 10^8}{2.56} = 77.3 \text{ MHz}.$$

The measured notch frequency is $f_0 = 59$ MHz. The velocity factor corresponding to the measured notch frequency follows from Eq. 8:

$$\mathrm{VF} = \frac{4 L f_0}{c} = \frac{4 \times 0.64 \times 59 \times 10^6}{3 \times 10^8} = 0.50.$$

Difference between calculated and measured f_0:

$$77.3 - 59 = 18.3 \text{ MHz} \quad \left(\frac{18.3}{77.3} \times 100 \approx 23.7\% \right).$$

9 Results

Quantity	Symbol	Measured/Calculated value
Filter type	–	Low-pass
Cutoff frequency (−3 dB)	f_c	540 MHz
Stopband frequency	f_s	Not determined (the manual gives no attenuation criterion)
Band-pass/notch frequencies	f_1, f_0, f_2	Not applicable (low-pass filter)
Filter insertion loss	IL	0.80 dB at 50 MHz (minimum measured); 0.80–2.81 dB over 50–500 MHz; maximum 42 dB at 740 MHz
Filter reflection coefficient	$\lvert\Gamma\rvert$	0.0264
Filter return loss	RL	31.56 dB
Stub notch frequency, calculated	f_0	77.3 MHz (VF = 0.66, L = 64 cm)
Stub notch frequency, measured	f_0	59 MHz
Velocity factor from measured f_0	VF	0.50

Table 4: Summary of the Part A results.

10 Conclusions

- **Filter type and response.** The insertion loss is small and slowly varying at low frequency (0.80 dB at 50 MHz, rising to 2.81 dB at 500 MHz) and increases steeply above about 540 MHz, reaching 20 dB at 650 MHz and a maximum of 42 dB at 740 MHz. The filter therefore passes frequencies below its cutoff and attenuates higher frequencies, i.e. it is a **low-pass filter**. The −3 dB cutoff frequency relative to the minimum passband insertion loss is f_c = 540 MHz, so the passband extends up to 540 MHz (the lowest measured frequency being 50 MHz). Beyond 740 MHz the insertion loss decreases again to 38 dB (770–800 MHz), 34 dB (850 MHz) and 36 dB (900 MHz).
- **Reflection coefficient and return loss.** In the flat passband (60–65 MHz) the ripple at the Tee junction is small ($V_{\max}$ = 680 mV, $V_{\min}$ = 645 mV), giving $\lvert\Gamma\rvert$ = 0.0264 and RL = 31.56 dB. About 2.6% of the incident voltage amplitude is reflected, so the filter input is well matched to the 50 Ω system.
- **Stub notch.** The quarter-wave open-circuit stub of length 64 cm gives a calculated notch frequency of 77.3 MHz for VF = 0.66, while the measured notch is at 59 MHz, lower by 18.3 MHz (23.7%). The measured value corresponds to VF = 0.50. The manual attributes differences between calculated and measured f_0 to the connector length and the open-end effect.
- **Sources of measurement uncertainty.**
  - In the Tee-junction method, cable loss, connector mismatch and oscilloscope loading are not removed by calibration, so $\lvert\Gamma\rvert$ and RL are approximate.
  - $\lvert\Gamma\rvert$ is obtained from a single $V_{\max}/V_{\min}$ pair, and $V_{\max}$, $V_{\min}$ must be the same voltage quantity.
  - The frequency step of the transmission sweep (10 MHz near the cutoff) limits the resolution of f_c.
  - The connector length and open-end effect affect the measured stub notch frequency.